Student Performance Prediction Using Machine Learning

Project Report

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Project Type: Beginner Machine Learning Project
Dataset: UCI Student Performance Dataset
Programming Language: Python

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1. Project Overview

This project focuses on predicting students' final academic grades using machine learning. A real-world student performance dataset from the UCI Machine Learning Repository was used for the analysis. Python and machine learning techniques were used to prepare the data, train prediction models, evaluate their performance, and make a prediction for a new student.

2. Objective

The main objective of this project is to develop a machine learning model that can predict students' final grades based on selected academic, personal, and lifestyle-related factors. The project also aims to understand the basic workflow of a machine learning problem, including data selection, feature selection, model training, prediction, and model evaluation.

3. Dataset

This project uses the Student Performance dataset obtained from the UCI Machine Learning Repository. The dataset contains information about students from two Portuguese schools and includes academic, demographic, family, and lifestyle-related factors. It contains 649 student records and 30 input features in the version used for this project. The target variable is G3, which represents the students' final grades. For this project, selected numerical features were used to train and evaluate Linear Regression models.

4. Tools and Technologies

The following tools and technologies were used to complete this project:

- **Python:** Used as the main programming language.
- **Pandas:** Used to select and organize the dataset features.
- **Scikit-learn:** Used to split the data, train Linear Regression models, generate predictions, and evaluate model performance.
- **Matplotlib:** Used to create the Actual vs. Predicted Grades scatter plot.
- **Google Colab:** Used as the cloud-based environment to write and execute Python code.

5. Methodology

5.1 Dataset Loading and Feature Selection

The Student Performance dataset was obtained from the UCI Machine Learning Repository using the ucimlrepo library. Pandas was used to select the required numerical features. The target variable, G3, represents the students' final grades.

5.2 Train-Test Split

The dataset was divided into training and testing sets using Scikit-learn's train_test_split function. A test size of 20% was used, with the remaining 80% used for training. The random state was set to 42 to make the split reproducible.

5.3 Model Development

Two Linear Regression models were developed. Model 1 used three features: study time, past failures, and absences. Model 2 used twelve numerical features, including study time, past failures, absences, parents' education levels, travel time, and selected family and lifestyle factors. Both models were trained using the training data.

5.4 Prediction and Evaluation

The trained models were used to predict final grades for the test data. Their performance was evaluated using Mean Absolute Error (MAE) and the R^2 score. An Actual vs. Predicted scatter plot was also created to visualize the predictions.

6. Model Evaluation and Results

6.1 Evaluation Metrics

The performance of both Linear Regression models was evaluated using Mean Absolute Error (MAE) and the coefficient of determination (R^2 score). MAE measures the average absolute difference between actual and predicted grades, where a lower value indicates smaller prediction errors. The R^2 score indicates how well the model explains variations in the target variable. A higher R^2 score generally indicates better performance.

6.2 Comparison of Model Performance

Model	MAE	R^2 Score
Model 1	2.2060	0.1496
Model 2	2.1332	0.1722

6.3 Interpretation of Results

Although Model 2 performed slightly better than Model 1, both models had limited predictive power. The low R^2 scores suggest that the selected features explained only a small portion of the variation in students' final grades. This highlights the importance of considering additional relevant factors and exploring other modelling approaches in future work.

7. Actual vs Predicted Graph

An Actual vs Predicted scatter plot was created to compare the actual final grades with the grades predicted by Model 2. Each point represents a student in the test dataset. The graph helps visualize how closely the predicted grades match the actual grades. Points closer to the ideal diagonal pattern indicate more accurate predictions, while points farther away represent larger prediction errors.

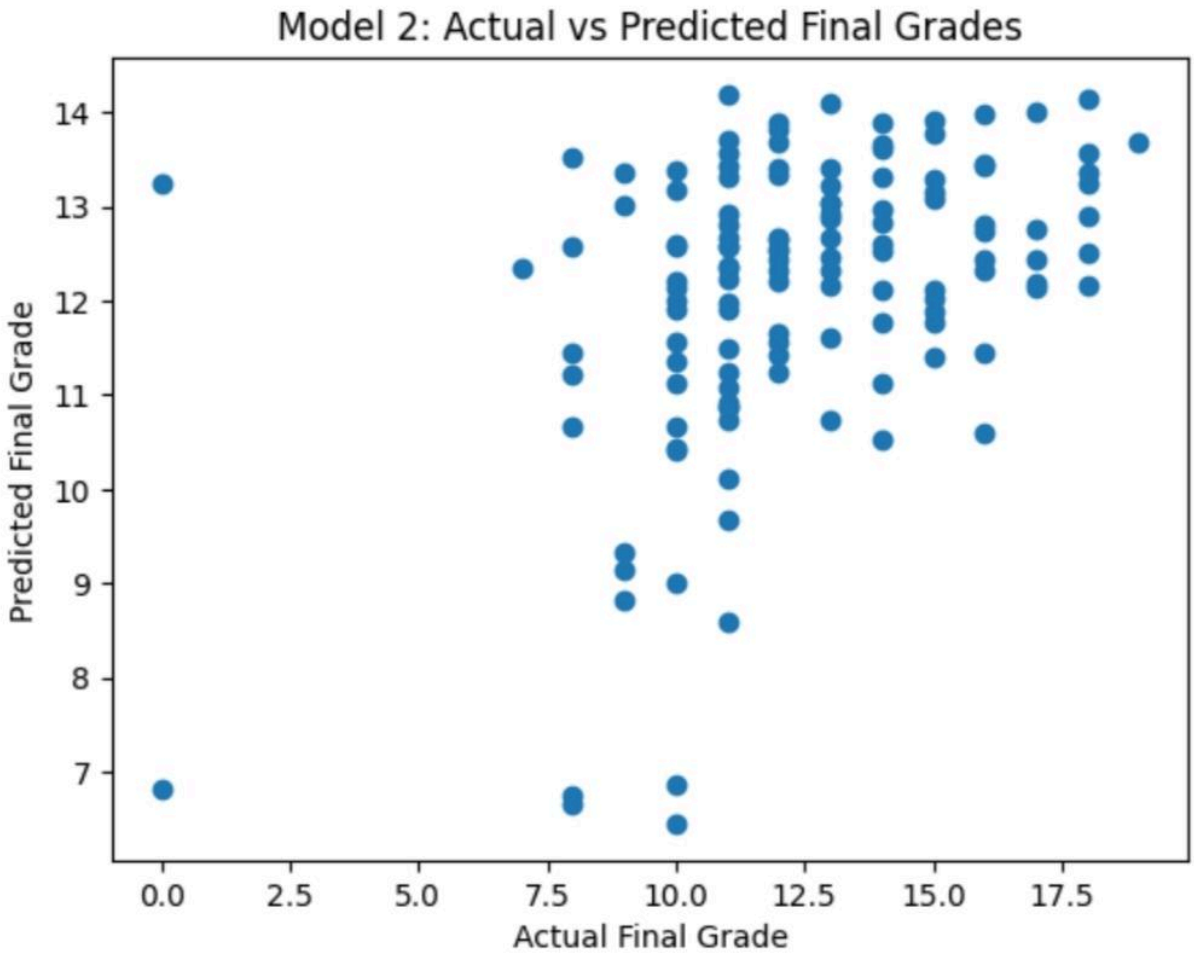


Figure 1: Actual vs Predicted Final Grades Using Model 2

8. New Student Prediction

A new prediction was generated using Model 2 for a hypothetical student with selected academic, family, and lifestyle-related characteristics. The model predicted a final grade of approximately 13.61 out of 20. This example demonstrates how a trained machine learning model can use input features to estimate a student's final grade. The prediction is illustrative and should not be considered an exact or guaranteed result.

Example Input Features

- Study Time: 3
- Past Failures: 0
- Absences: 4
- Mother's Education: 3
- Father's Education: 3
- Travel Time: 1
- Family Relationship: 4
- Free Time: 3
- Going Out: 2
- Workday Alcohol Consumption: 1
- Weekend Alcohol Consumption: 1
- Health: 4

9. Conclusion

This project provided practical experience in building and evaluating a machine learning model using Python. Two Linear Regression models were developed to predict students' final grades using selected academic, family, and lifestyle-related features. Model 2 performed slightly better than Model 1, although both models had limited predictive power. The project helped develop an understanding of dataset preparation, feature selection, model training, prediction, and performance evaluation. It also highlighted the importance of interpreting machine learning results carefully and recognizing the limitations of a model. This project serves as a foundation for further learning and developing more advanced machine learning projects.

10. Limitations

This project has several limitations. First, the models used only selected numerical features from the dataset, so other factors that may influence students' grades were not included. Second, both models achieved relatively low R^2 scores, indicating that they explained only a small portion of the variation in final grades. Third, the dataset contains student records from two Portuguese schools, so the results may not apply directly to students in other educational systems or countries. Finally, the model's predictions are estimates and should not be treated as exact results. Further work could explore additional features, other machine learning algorithms, and more diverse datasets to improve the model's performance.

11. References

Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Dataset Source: UCI Machine Learning Repository — Student Performance Dataset https://archive.ics.uci.edu/dataset/320/student%2Bperformance